31.12.2023 and 11.02.2024

8.struct node{

```
  int value;
  struct*next;
}void rearrange(struct node*list){
   struct node*p,*q;int temp;
   if(!list || !list->next)
     return;
  p=list;q=list
  while(q){
   temp=p->values;
   p->value=q->value;
   q->value=temp;
   p=q->next;
   q=p?p->bext:o;
}
}
```

list containing integers 1,2,3,4,5,6,7

output-2,1,4,3,6,5,7

4.which sorting technique out of the following options uses the last number of comparisions to sort the

array[12,32,43,56,65,76,87,98] in ascending order ?

a.selection sort

b.merge sort

c.quicksort using the last pivot

d.insertion sort

answer-b. merge sort.

5.the conatenation of two lists list to be performed 0(1)time which of the following imprementations of a list should be

used?

a.singley linkedlist

b.doubley linkedlisst

c.circular doubley linked list

d.array impleamention of list

answer-c. circular doubly linked list.

1.program preads in 500 integers in the range[0..100]representing the scores of 500 students.if then prints the frequency of each score

above 50.what would be the best way for p to store the frequencies?

answer like- int scores[500];

2.in the balanced binary tree in the below figure .how many nodes will become unbalanced when q node is inserted as a

child of the node g?

```
    a
  /   \
 b     e
/ \   /
c  d  f
/
g
```

answer-2 node

3.suppose a circular queue of capcatiy (n-1) elementss is implemented with an array of n elements.Assume that the insertion and selection operation are carried out using rear and front as array index variables.Initaially ,REAR=FRONT=0.

the condition to detect queue fulll and queeue empty are

answer -b.(REAR+1)modn=FRONT.

9.The following function compute x^y (x to the power of y) for postive integers x and y.

```
int exp(int x,int y){
  int res=1,a=x,b=y;
  while(b!=0){
  if(b%2==0){
   a=a*a;
   b=b/2;
}
else{
    res=res*a;
    b=b-1;
}
}
return res;
}
```

a.x^y=a^b

b(res*a)^y=(res*X)^b

c.x^y=res^x a^b

d.x^y=(res*a)^b

answer -d.x^y=(res*a)^b

6.for the below c snippet,find f(5).

```
int f(int n){
static int r=0;
if(n<=0)return 1;
if(n>3){
r=n;
return f(n-2)+r;
}
}
```

Answer-9.

7.suppose you are given implementation of a queue of integers.the operations that can be performed on the

queue are:

isempty(Q)->returns true if the queue q is empty false otherwise.

delete(Q)->deletes the element at the front of the queue q and returns its value.

insert(Q,i)->inserts the integers i at the near of the queue.

```
void f(queue Q){
 int i;
 if(!isempty(Q)){
 i=delete(Q);
 f(Q);
insert(Q,i);
}
}
```

the function f is called with the queue containing integers 8,1,6,3,5,4,7 in the given order what will be the contents of the queue after the function completes?

answer-[7, 4, 5, 3, 6, 1, 8].

10.Assume that swap(&x,&y)contents of x and y.

```
int main(){
int array[]={3,5,1,4,6,2};
int done=0;
int i;
while(done==0){
done=1;
for(i=0;i<=4;i++){
if(array[i]=1;i--){
if(array[i]>array[i-1]){
swap(&array[i],&array[i-1]);
done=0;
}
}
}
print("%d",array[3]);
}
}
```

answer-no output

ZOHO 27-04-2024

9. // Online C compiler to run C program online

```
#include <stdio.h>
```

```
int main(){
    int c[]={2,3,4,6,5};
    int j,*p=c,*q=c;
    for(int j=0;j<5;j++){
        printf("%d",*c);
        ++q;
    }
    for(j=0;j<5;j++){
        printf("%d",*p);
        ++p;
    }
    return 0;
}
```

Answer-2222223465

10. // Online C compiler to run C program online

```
#include <stdio.h>

void main(){
    int n=12,res=1;
    while(n>3){
        n-=3;
        res*=3;
    }
    printf("%d",n*res);
}
```

Answer-81

7. // Online C compiler to run C program online

```
#include <stdio.h>

void main(){
    int i=0,j,k,n=4;
    while(i++<n){
        if(i%2==1){
            k=1;
        }else{
            k=0;
        }
        for(j=1;j<=i;j++,k+=1){
            printf("%d",k);
        }
        printf("\n");
    }
}
```

Answer-

1

01

123

0123

4. // Online C compiler to run C program online

```
#include <stdio.h>
int compare(int a,int b,int c){
    if(a-b>0 && a-c>0){
        return c;
```

```
    }else{
        if(b-c>0){
            return a;
        }else{
            return b;
        }
    }
}
int main(){
    printf("%d",
    compare(compare(88,89,91)
    ,compare(90,41,17),
    compare(75,100,96)));
    return 0;
}
```

Answer-75

5. // Online C compiler to run C program online

```
#include <stdio.h>
int main(){
    int n,x,count,i;
    n=68;
    for(x=1;x<=n;x++){
        if(n%x==0){
            for(count=0,i=1;i<=x;i++){
                if(x%i==0){
                    count++;
                }if(count==2){
```

```
                printf("\t%d",x);
            }
        }
    }
  }
  return 0;
}
```

Answer- 2 4 4 17 34 34 34 34 34 34 34 34 34 34 34 34 34 34 34 68 68

6. // Online C compiler to run C program online

```
#include <stdio.h>
int main(){
  int n=4,i,j,sum=0;
  int a[][4]={
   {12,15,16,20},{4,10,2,21},{33,14,23,9},{22,43,54,6}
  };
  for(i=0;i<n;i++){
    for(j=0;j<n;j++){
      if(i==j){
        sum=sum+a[i][j];
      }
    }
  }
  printf("%d",sum);
  return 0;
}
```

Answer-51

1. // Online C compiler to run C program online

```
#include <stdio.h>
int main(){
    int a,b,number,result,i;
    a=6;b=14;
    number=(a>=b)?a:b;
    for(i=number;;i++){
        if(i%a==0 && i%b==0){
            result=i;
            break;
        }
    }
    printf("%d",result);
    return 0;
}
```

Answer-42

2. // Online C compiler to run C program online

```
#include <stdio.h>
int main(){
    char *x="Zoho123Corp45";
    int result=0,i;
    for(i=0;x[0]!='O';i++){
        if(x[i]>=48&& x[i]<=57){
            result=result+(x[i]='O');
        }

    }
    printf("%d",result);
```

```
    return 0;
}
```

Answer-Segmentation fault

3. // Online C compiler to run C program online

```
#include <stdio.h>
int i=0;
int fun(int a){
    i++;
    if(a>99){
        return a-13-1;
        return fun(fun(a+14+11));
    }
}
void main(){
    printf("%d",fun(69));
    printf("%d",i);
}
```

11